

Authors' reply

We thank Marcel Koenigkam Santos and colleagues for their thoughtful comments on our Review.¹ We agree that interstitial lung disease (ILD) with airway-centered fibrosis represents a major diagnostic challenge, and that patients with this pattern often have hypersensitivity pneumonitis,² gastro-oesophageal reflux, aspiration, or connective tissue disease.³ Given these multiple associations, the prevalence of idiopathic airway-centered fibrosis⁴ remains unclear. The ATS/ERS classification paper in 2013 recognised bronchiolocentric patterns of interstitial pneumonia as a rare histological finding, but it was not clear whether this finding could be idiopathic.⁵ Previous descriptions of the CT findings in this entity have been quite variable, including subpleural consolidation,⁶ subpleural interstitial abnormality,⁷ peribronchovascular ground glass abnormality,³ and peribronchovascular interstitial thickening and traction bronchiectasis.⁸ Because the CT findings are relatively non-specific, most of these cases would currently be included in the category of CT pattern indeterminate for UIP. The lack of a consistent, clear histological definition of airway-centered fibrosis has hampered understanding of this entity. Therefore, we agree with the authors' suggestion that a systematic review of this topic is appropriate.

With regard to the images presented in our online repository of high-resolution CT scans showing features of UIP, we hope to implement some of the suggestions of Santos and colleagues as we continue to update these cases.

We also appreciate the comments of Lee Fidler and Shane Shapera. The reason for including the category of CT pattern indeterminate for UIP as a separate entity is because we believe that the probability

of histological usual interstitial pneumonia (UIP) in this group is substantially higher than in patients with CT features most consistent with a non-IPF diagnosis. Although the paper⁹ by Chung and colleagues did not support this contention, the population in this retrospective study might have been biased by the inclusion of patients enrolled in clinical IPF trials and the requirement for all patients to have had a surgical lung biopsy. Additional studies with broader populations are needed to confirm the prevalence of an IPF clinical diagnosis for patients in the indeterminate imaging category and the consistent with non-IPF diagnosis imaging category.

We defined the category CT pattern indeterminate for UIP as having evidence of fibrosis with some inconspicuous features suggestive of non-UIP pattern. In most instances, these cases demonstrate some combination of ground glass opacity, lobular mosaic perfusion, and lobular air trapping, associated with evidence of fibrosis. We agree with Fidler and Shapera that use of this definition is challenging given the absence of clear and reproducible descriptions of what features qualify as inconspicuous. We hope that the publication of these guidelines and use of the online image repository will provide an impetus to study the specificity of CT features that suggest a non-IPF alternative diagnosis, thus providing a sharper delineation between the recommended diagnostic categories.

Competing interests are the same as in the original Review.

**David A Lynch, Nicola Sverzellati, William D Travis, Thomas V Colby, Yoshikazu Inoue, Andrew G Nicholson, Suhail Raoof, Luca Richeldi, Christopher J Ryerson, Jay H Ryu*
lynchd@njhealth.org

Department of Radiology, National Jewish Health, Denver, 80206 CO, USA (DAL); Department of Medicine and Surgery, University of Parma, Parma, Italy (NS); Department of Pathology, Memorial Sloan Kettering Cancer Center, New York, NY, USA (WDT); Department of Pathology, Mayo Clinic Scottsdale, Scottsdale, AZ, USA (TVC); Department of Histopathology, Royal Brompton and Hospital NHS Foundation Trust and National Heart and Lung Institute, Imperial College, London, UK (AGN); Clinical Research Center, National Hospital Organization Kinki-Chuo Chest Medical Center, Osaka, Japan (YI); Division of Pulmonary Medicine, Lenox Hill Hospital, New York, NY, USA (SR); Unità Operativa Complessa di Pneumologia, Agostino Gemelli University Hospital of the Catholic University of the Sacred Heart, Rome, Italy (LR); Department of Medicine, University of British Columbia and St Paul's Hospital, Vancouver, BC, Canada (CJR); and Division of Pulmonary and Critical Care Medicine, Mayo Clinic Rochester, Rochester, MN, USA (JHR)

- 1 Lynch DA, Sverzellati N, Travis WD, et al. Diagnostic criteria for idiopathic pulmonary fibrosis: a Fleischner Society White Paper. *Lancet Respir Med* 2018; **6**: 138–53.
- 2 Wang P, Jones KD, Urisman A, et al. Pathologic findings and prognosis in a large prospective cohort of chronic hypersensitivity pneumonitis. *Chest* 2017; **152**: 502–09.
- 3 Kuranishi LT, Leslie KO, Ferreira RG, et al. Airway-centered interstitial fibrosis: etiology, clinical findings and prognosis. *Respir Res* 2015; **16**: 55.
- 4 Yousem SA, Dacic S. Idiopathic bronchiolocentric interstitial pneumonia. *Mod Pathol* 2002; **15**: 1148–53.
- 5 Travis WD, Costabel U, Hansell DM, et al. An official American Thoracic Society/European Respiratory Society statement: update of the international multidisciplinary classification of the idiopathic interstitial pneumonias. *Am J Respir Crit Care Med* 2013; **188**: 733–48.
- 6 de Carvalho ME, Kairalla RA, Capelozzi VL, Deheinzelin D, do Nascimento Saldiva PH, de Carvalho CR. Centrilobular fibrosis: a novel histological pattern of idiopathic interstitial pneumonia. *Pathol Res Pract* 2002; **198**: 577–83.
- 7 Mark EJ, Ruangchira-urai R. Bronchiolitis interstitial pneumonitis: a pathologic study of 31 lung biopsies with features intermediate between bronchiolitis obliterans organizing pneumonia and usual interstitial pneumonitis, with clinical correlation. *Ann Diagn Pathol* 2008; **12**: 171–80.
- 8 Churg A, Myers J, Suarez T, et al. Airway-centered interstitial fibrosis: a distinct form of aggressive diffuse lung disease. *Am J Surg Pathol* 2004; **28**: 62–68.
- 9 Chung JH, Chawla A, Peljto AL, et al. CT scan findings of probable usual interstitial pneumonitis have a high predictive value for histologic usual interstitial pneumonitis. *Chest* 2015; **147**: 450–59.

For examples of the CT features of UIP see <http://get.pacsbin.com/fleischner/>